

종격동 질환, 종양

흉부외과 채민철

Contents

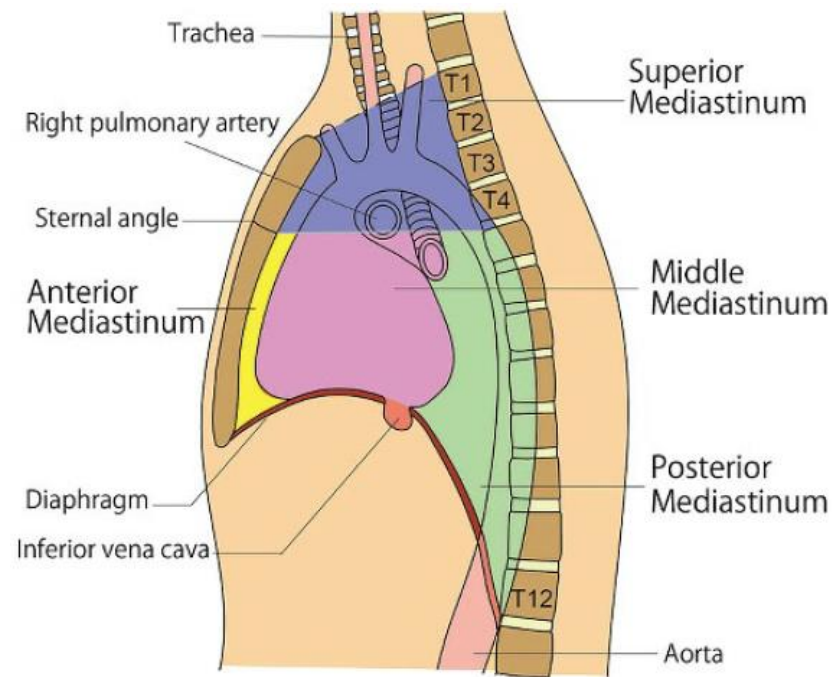
- Mediastinal disease
 - Pneumomediastinum
 - Mediastinal infection
 - Mediastinal tumor

Mediastinum

Mediastinum

 폐와 폐. 사이 공간

- Thoracic space located between the two pleural cavities
- Boundary
 - Superior : Thoracic inlet
aortic arch, trachea 갈라지는 bifurcation 정도 level 기준
 - Inferior : Diaphragm
 - Anterior : Sternum
pericardium 기준
 - Posterior : Thoracic vertebral column



Pneumomediastinum

Pneumomediastinum

종격동에 공기가 보임

- Consists of air or other gas in the mediastinum

대부분 공기가 지나가는 통로에 대한 평가

Table 84-1 Origins of Air in the Mediastinum

UPPER RESPIRATORY TRACT

Head and neck infection (odontogenic, salivary glands, cervical adenitis, tonsillitis, peritonsillar abscess, osteomyelitis of facial bones)
Fractures (involving paranasal sinuses, orbit, mandibles, other facial bones)
Other mucosal disruption (trauma, surgery, attempted endotracheal intubation)
Dental procedures (extractions, air-turbine drilling)

INTRATHORACIC AIRWAYS

Blunt or penetrating chest trauma
Foreign body
Iatrogenic (bronchoscopy, bronchial brushing, transbronchial biopsy, needle aspiration)
Neoplasm

LUNG PARENCHYMA

Direct disruption of alveoli (penetrating trauma, surgery, transbronchial biopsy, needle aspiration)
"Spontaneous" alveolar rupture (between alveolus and adjacent bronchovascular bundle)

GASTROINTESTINAL TRACT

Esophageal perforation
Via pneumoperitoneum or pneumoretroperitoneum (gastric or intestinal perforation, diverticulitis, pneumatosis cystoides intestinalis, endoscopy, biopsy, infection)

EXTERNAL TO THE BODY

Penetrating trauma to neck or chest
Surgical procedures (tracheotomy, mediastinoscopy, sternotomy)
Via subcutaneous emphysema in association with chest tube insertion

INFECTION WITH GAS-PRODUCING ORGANISMS

Acute bacterial mediastinitis
Head and neck infections

Pneumomediastinum

Pneumomediastinum

- **Spontaneous** pneumomediastinum **self limited**
 - Without obvious etiology
 - Predisposing condition
 - Young adult, benign, self limited
- **Secondary** pneumomediastinum
 - Specific pathologic condition
 - Higher morbidity, mortality

Infection으로 발전할 수 있음

Spontaneous	Secondary
VOLUNTARY ALTERATION IN BREATHING PATTERN Marijuana smoking and cocaine inhalation Pulmonary function testing Mountain climbing—"voluntary pressure breathing" Wind instrument playing Yelling, shouting, and singing	BAROTRAUMA Mechanical ventilation Noninvasive mechanical ventilation Manual ventilation during resuscitation, anesthesia, or transport Equipment malfunction or misconnection in anesthesia or oxygen therapy Heimlich maneuver Deceleration injury
INVOLUNTARY ALTERATION IN BREATHING PATTERN Childbirth Vomiting Seizures; status epilepticus Violent coughing; sneezing; hiccupping Heavy lifting; athletic competition Straining at stool	INVASIVE PROCEDURES Esophageal perforation Endobronchial needle aspiration
INTRINSIC AIRWAY AND LUNG DISEASE Asthma Atelectasis Bronchiolitis Pneumonia Influenza Measles Tuberculosis Silicosis Foreign body Tumor	

Pneumomediastinum

- Clinical presentation

- Symptom

- Chest pain, aggravation by movement, breathing, position change
 - Dysphagia or dysphonia : neck, retropharyngeal, perilaryngeal air dissection
 - Dyspnea, cough 비특이적

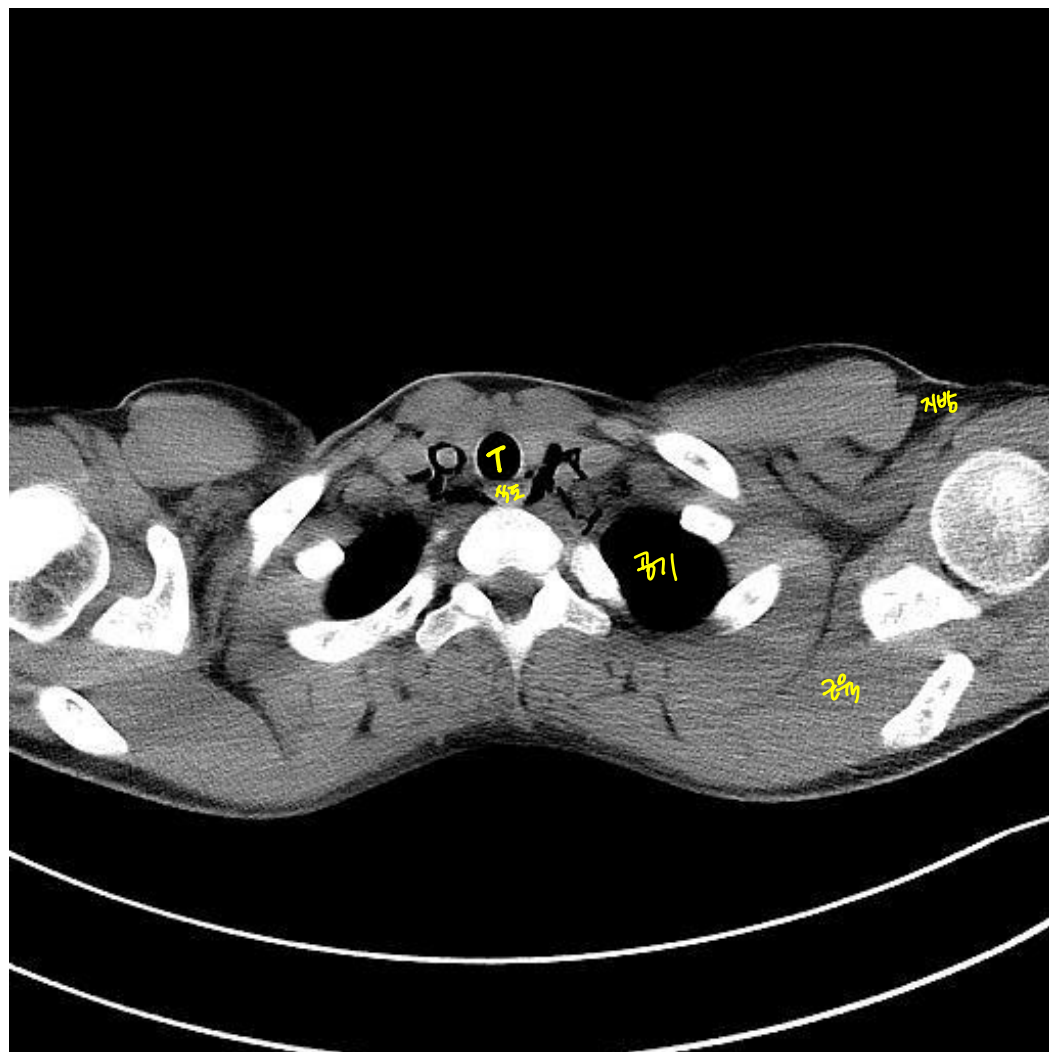
- Physical examination

- Palpable crepitation in neck, supraclavicular area, Hamman sign

- Treatment

경우에 따라서는 X-ray 에선 잘 안 보이고 CT에서 잘 보인 경우 있음

- Treatment of predisposing or precipitating factors
 - Oxygen supply
 - small infraclavicular venting incisions 양이 많은 경우-밖으로 공기 빼주려고
 - During positive pressure ventilation : reduction of TV, PEEP



조금씩 있는 것들은 만져서 느껴지지 않는데
양이 많으면 피부를 통해 만지면 느껴짐

Mediastinitis

Mediastinitis

위험한 infection

Table 84-3 Acute Mediastinitis: Etiologies and Clinical Settings

PERFORATION OF A THORACIC VISCUS

Esophagus 제일 많음

Forceful vomiting (**Boerhaave syndrome**) 복압-가슴전이-하부식도 전층 터짐-배에 있던 것들이 흉강으로 빠져나옴

Direct penetrating trauma

Impacted **foreign body** 음식, 알약

Instrumentation: esophagoscopy; sclerotherapy; esophageal obturator airway

Erosion: carcinoma; necrotizing infection

Trachea or Main Bronchi 흔치 않음

Direct penetrating trauma

Instrumentation: diagnostic bronchoscopy; therapeutic bronchoscopy; intubation

Foreign body

Erosion of carcinoma

DIRECT EXTENSION OF INFECTION FROM OUTSIDE MEDIASTINUM 목 쪽 infection이 타고 내려옴

Retropharyngeal space; odontogenic

Pancreatitis

Lung; pleura; pericardium

Lymph node

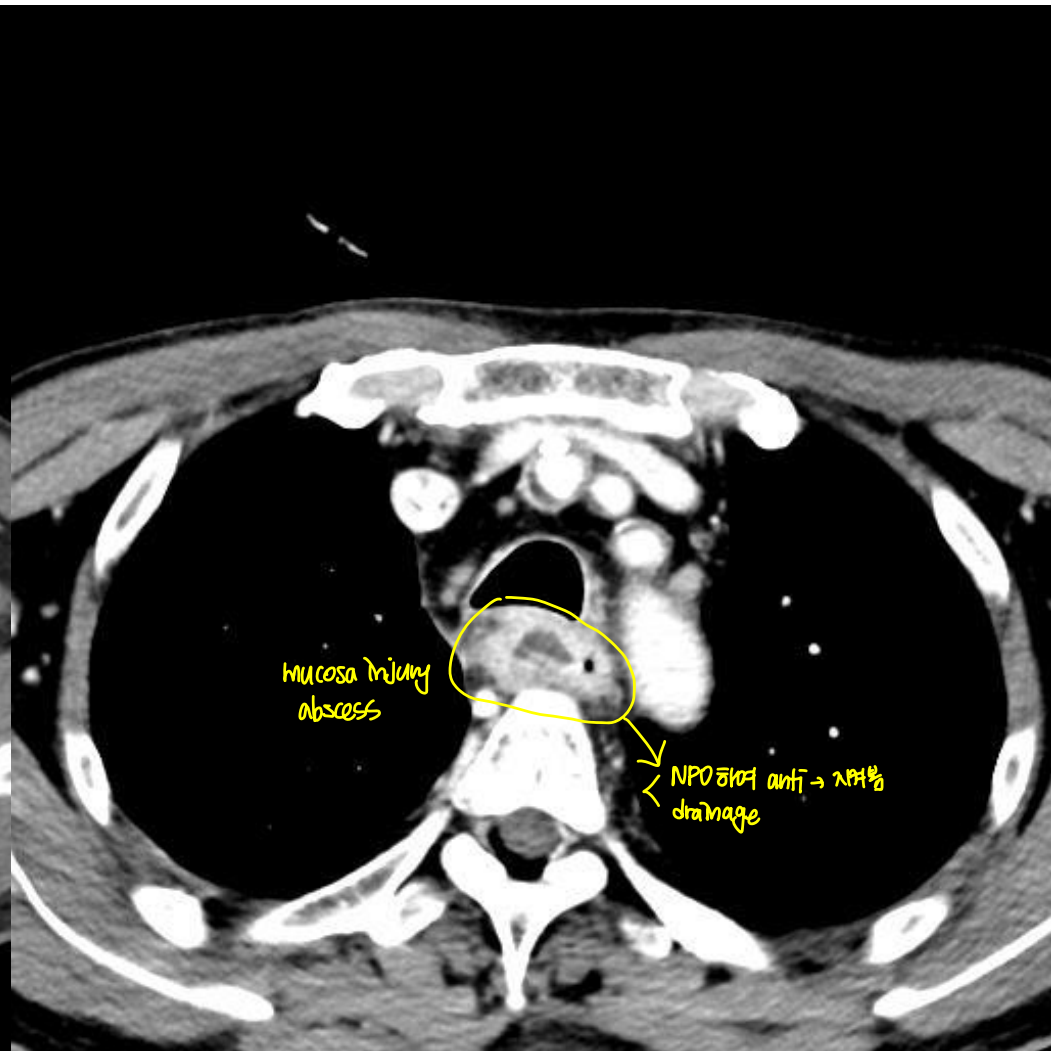
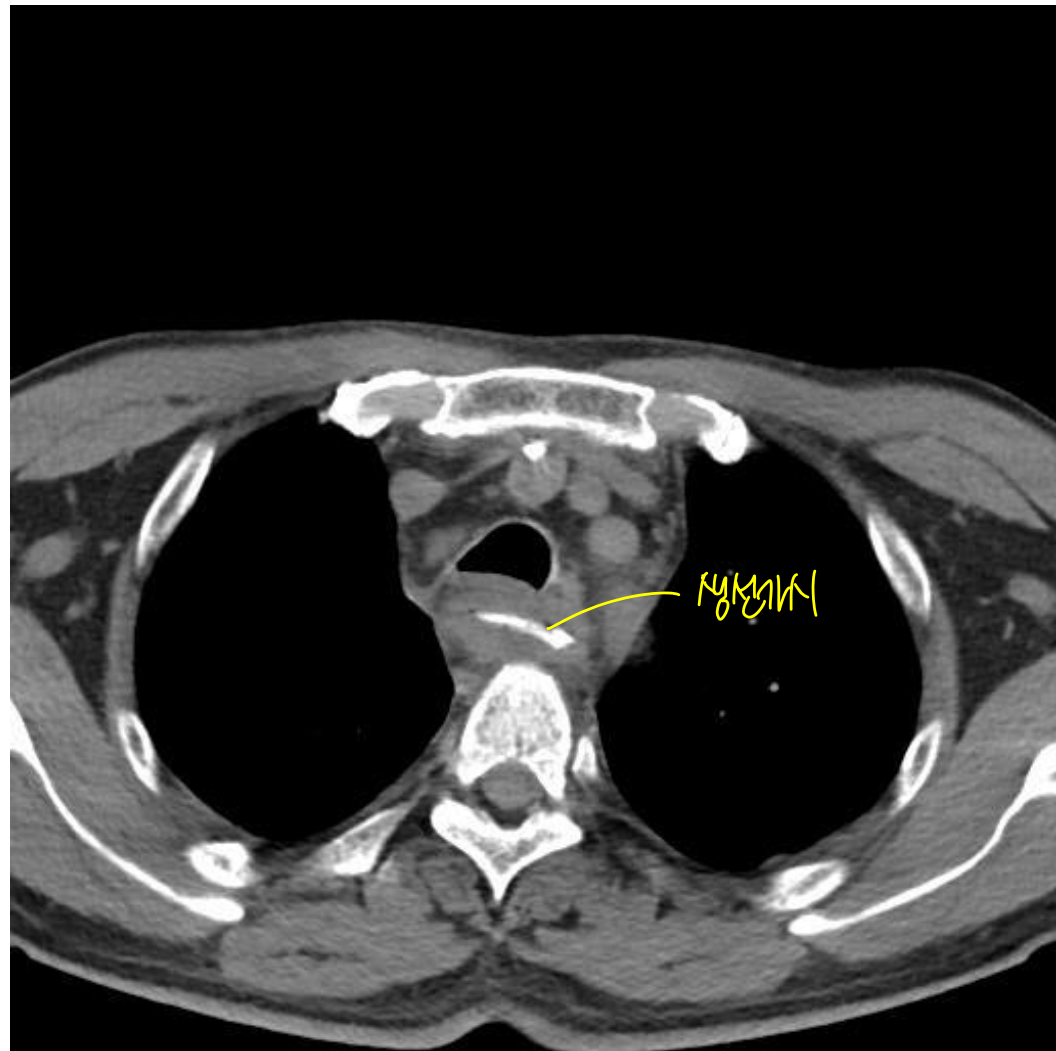
Paraspinous abscess

MEDIASTINITIS AFTER **CARDIOTHORACIC SURGERY**

INHALATIONAL ANTHRAX

Esophageal Perforation

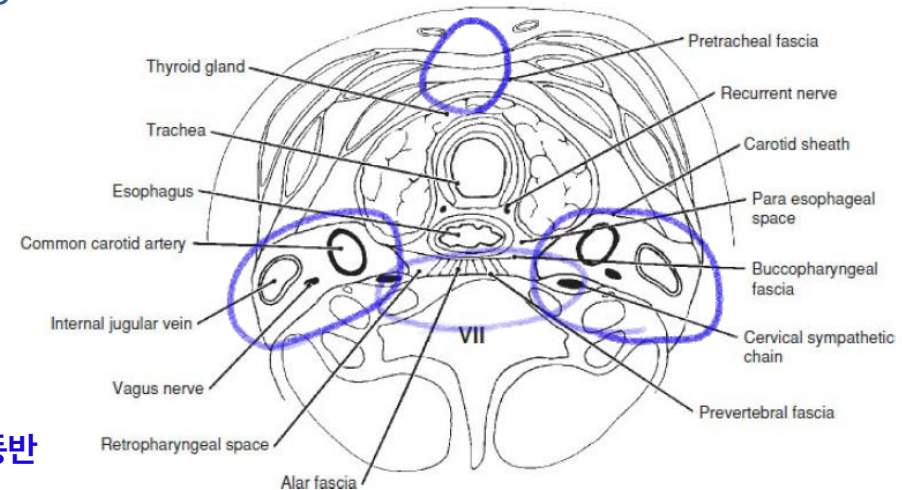
- m/c cause of posterior mediastinal abscess
- Treatment
 - Intravenous antibiotics
 - Surgical drainage and debridement
 - widely opening the mediastinal compartment by incision of the pleura
 - Débridement, irrigation, drainage, closure of the esophageal perforation



Descending necrotizing mediastinitis drainage 잘 하면 밑으로 내려올 일 없음

- Infection that has extended directly into the mediastinum from a head and neck source
 - Periodontal, peritonsillar, odontogenic, pharyngeal infection
 - 3 potential planes :
 - Pretracheal plane
 - Perivascular plane : carotid sheath
 - Prevertebral plane : retropharyngeal space
 - Mixed infection : aerobic / anaerobic
 - Rapid progression : sepsis, multiorgan failure

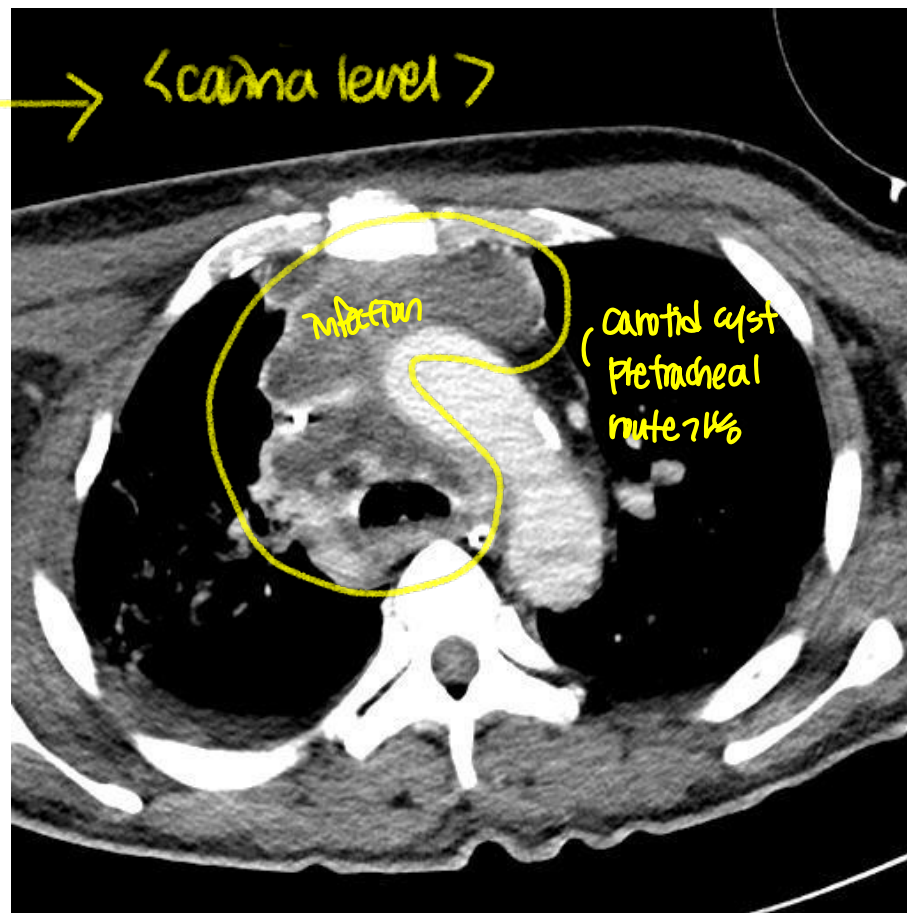
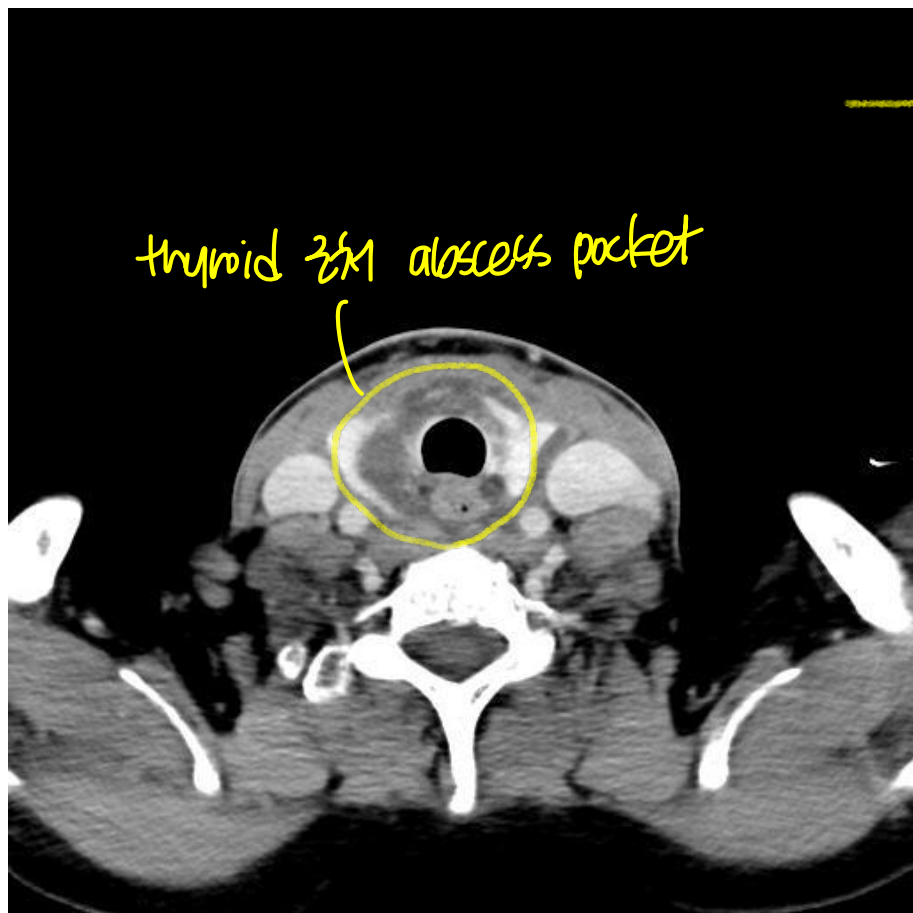
infection 계속 내려오면 -> sepsis로 잘 넘어가고 multiorgan failure 동반



※ 목에서 진단 빨리 내리고 빨리 처치하는 것이 좋음 ※

Descending necrotizing mediastinitis

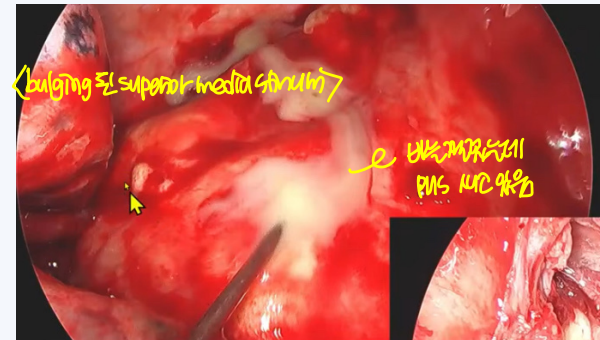
- Clinical manifestation
 - Under treatment for deep cervical infection
 - Symptom : stiffness, swelling, neck pain, cranial nerve deficits, trismus, stridor
 - Mediastinal involvement : 12hr – 2wks after onset of deep cervical infection
 - High mortality rate 하루이틀이면 충분
surgical drainage 필수적/한번으로 안되면 2~3번
 - Early recognition and diagnosis : important
- Treatment : Aggressive surgical drainage, Prompt antibiotic administration
 - Transcervical approach : superior mediastinum (above level of carina) 목에서 끝내는 것이 가장 좋음
 - Transthoracic approach 잘 하지는 않음



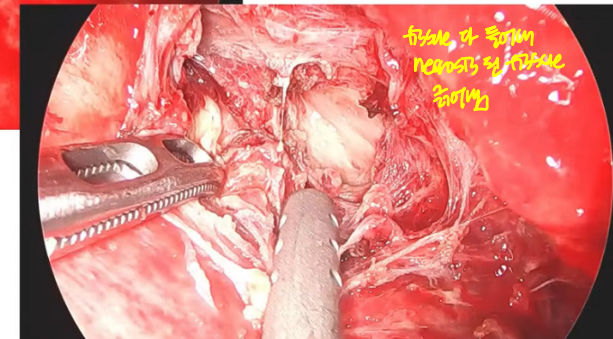
머리쪽으로 infection 올라가기도 함

Mass in the mediastinum

Anterior Mediastinum	Middle Mediastinum	Posterior Mediastinum
- Thymic neoplasms - Germ cell tumors - Teratoma - Seminoma - Nonseminomatous germ cell tumors - Embryonal cell carcinoma - Choriocarcinoma - Lymphoma - Hodgkin lymphoma - Non-Hodgkin lymphoma - Thyroid neoplasms - Parathyroid neoplasms - Mesenchymal tumors - Lipoma - Fibroma - Lymphangioma - Hemangioma - Mesothelioma - Sarcoma - Diaphragmatic hernia (Morgagni) - NUT midline carcinoma	- Lymphadenopathy - Reactive and granulomatous inflammation - Metastasis - Angiofollicular lymphoid hyperplasia (Castleman disease) - Lymphoma - Developmental cysts - Pericardial cyst - Foregut duplication cysts - Bronchogenic cyst - Enteric cyst - Others - Vascular enlargements - Diaphragmatic hernia (hiatal)	- Neurogenic tumors - Meningocele - Esophageal lesions - Carcinoma - Diverticula - Diaphragmatic hernia (Bochdalek) - Miscellaneous



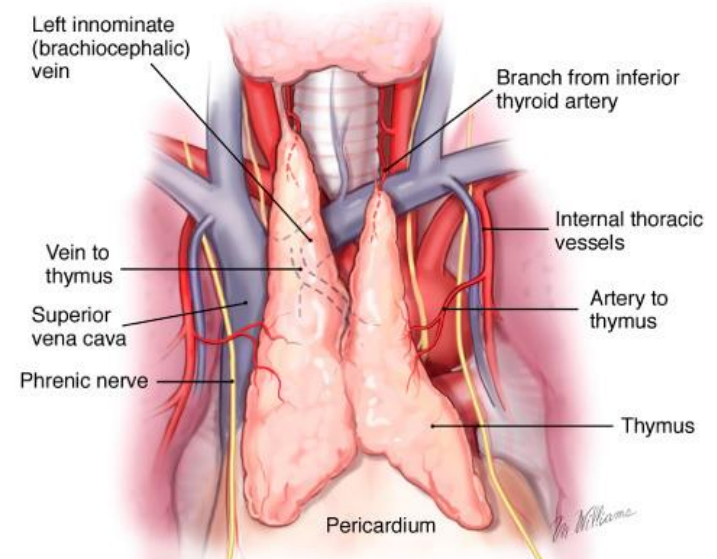
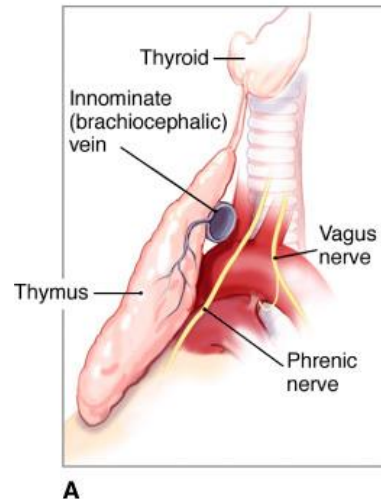
수술자체가 어렵지는 X, infection된 tissue들이어서 피가 많이 남
위쪽 혈관 구조를 박리 안되는 경우가 많음



Thymus

- Anatomy

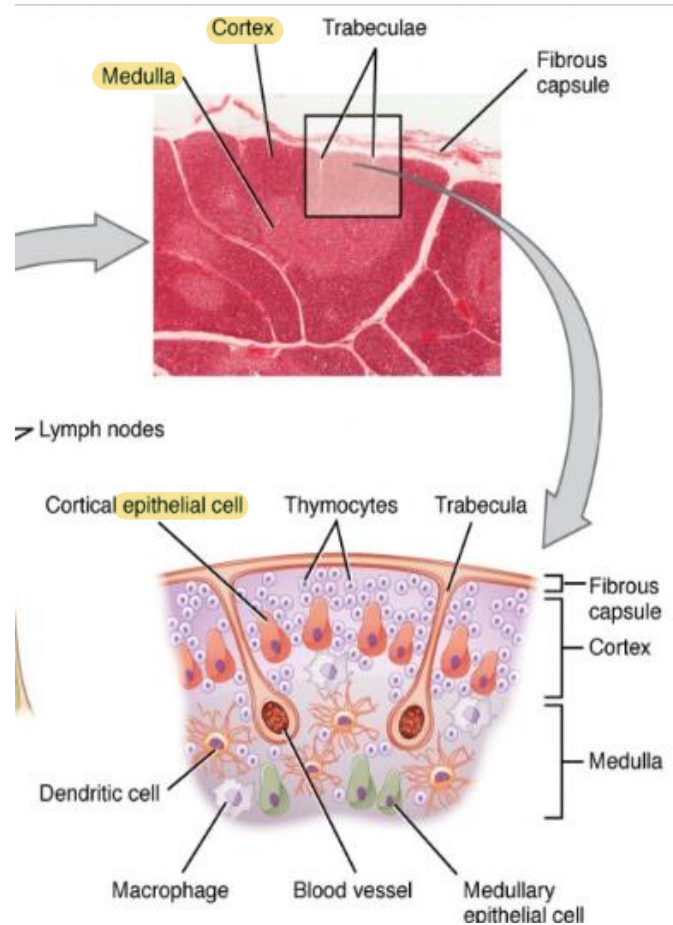
- Roughly H-shaped
- Upper pole : Attachment to thyroid gl. by thyrothymic lig.
- Lower pole : extend down over the pericardium
- Newborn :15g, Grow until puberty : 30-40g
- Gradual involution throughout adulthood, weight reduction 1-3%/yr, 20-28g **어릴 땐 크다 나이들며 점점 퇴화**



Thymus *T cell maturation* 과 관련 있는 조직

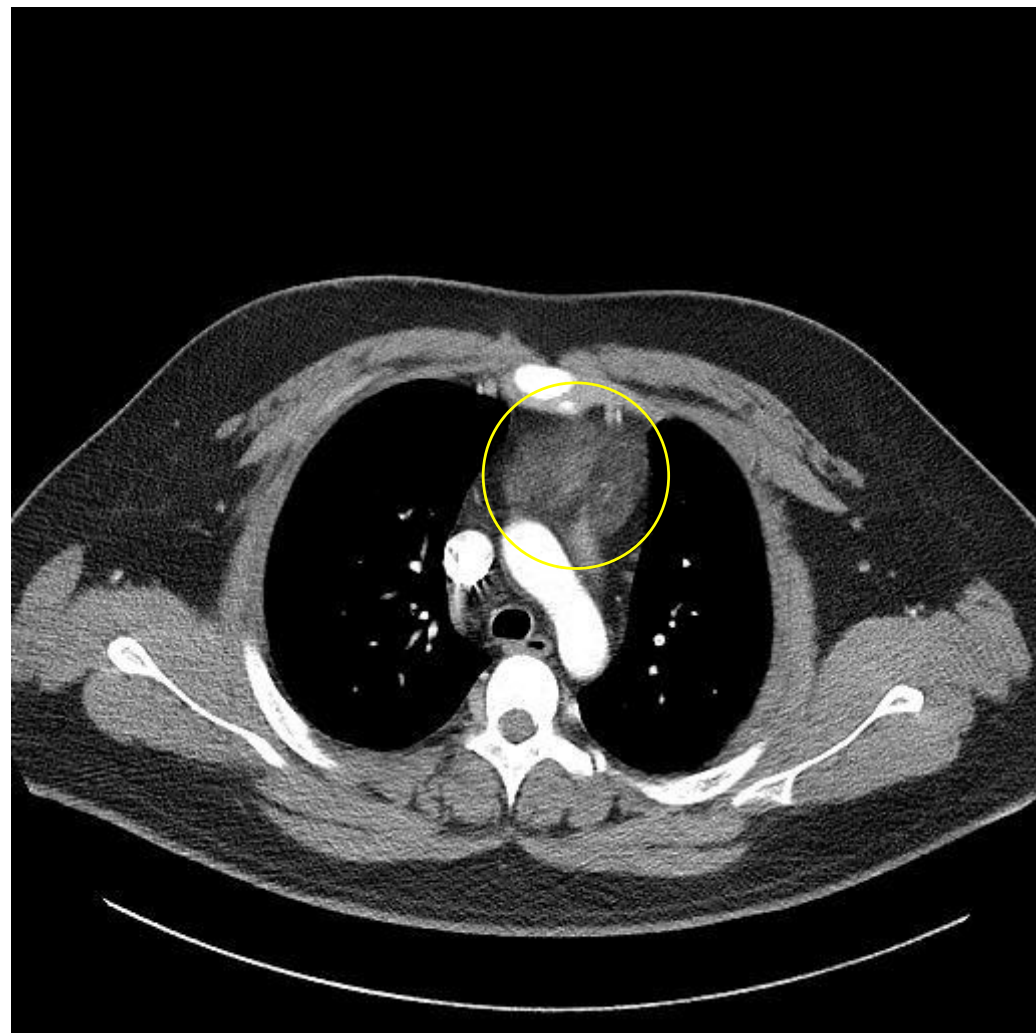
- Histology

- Fibrous capsule
- Fibrous connective tissue septa
- Lobule : cortex, medulla
- Thymic epithelial cell (TEC)
 - cTEC, mTEC
- Thymic lymphocyte
 - Prothymocyte
 - Immature T lymphocyte
 - Mature T lymphocyte



Thymic hyperaplasia 조직 비대

- Lymphoid or follicular hyperplasia 조직 좀 더 치밀
 - Presence of lymphoid follicles with activated germinal centers
 - Associated with autoimmune disease
 - 60-90% of myasthenic patients without thymoma
- True thymic hyperplasia 실제로 부피가 커지며 airway 누름
 - Increase in the size and weight of the thymus gland
 - Thymic rebound hyperplasia : associated with systemic disease, physical stress
 - Idiopathic true Thymic hyperplasia : rare, young age

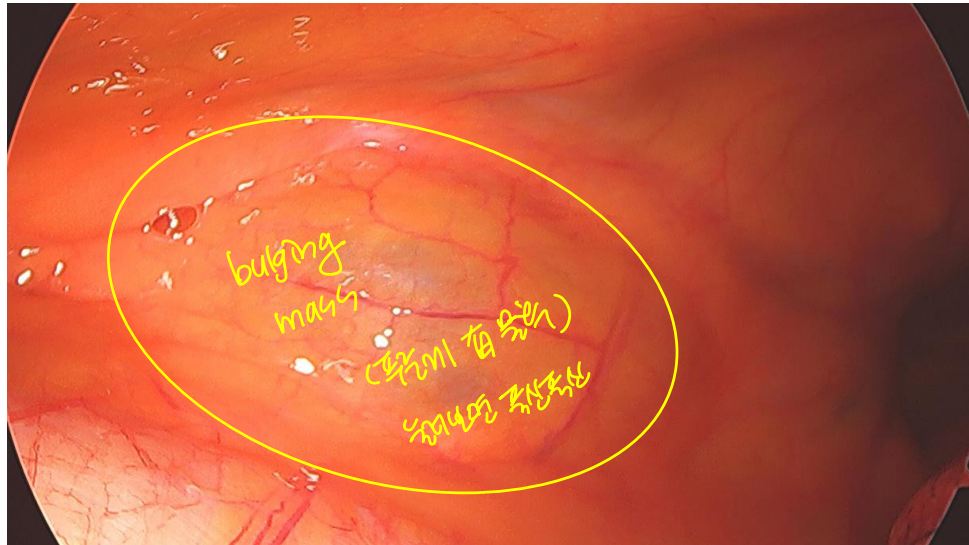


Thymic cyst 크기 다양

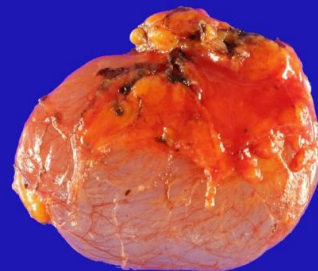
- Clinical presentation
- Asymptomatic, incidental finding
- Unilocular location 여러 개 잘 X
- Benign disease 크기 변화는 있을 수 있으나 malignant transformation은 드물다.
- Surgical excision (예전에 많이 함)
 - Definitive diagnosis, r/o other tumor (cystic thymoma)
 - Complete treatment

**infection, 크기가 너무 큰 경우 아니면 요즘에는 경과관찰

301자 앞의 301로 301제가 들어간 것
 V 301제 X
 MRI 301제 301제 301제



301제 301제 301제 301제 301제 301제



Mediastinal tumor

Thymic tumor

- Tumor arising from thymic epithelial and neuroendocrine cells

Thymoma

Type A thymoma, including atypical variant
 Type AB thymoma
 Type B1 thymoma
 Type B2 thymoma
 Type B3 thymoma
 Micronodular thymoma with lymphoid stroma
 Metaplastic thymoma
 Other rare thymomas
 Microscopic thymoma
 Sclerosing thymoma
 Lipofibroadenoma

Thymic carcinoma

Squamous cell carcinoma
 Basaloid carcinoma
 Mucoepidermoid carcinoma
 Lymphoepithelioma-like carcinoma
 Clear cell carcinoma
 Sarcomatoid carcinoma
 Adenocarcinomas
 Papillary adenocarcinoma
 Thymic carcinoma with adenoid cystic carcinoma-like features
 Mucinous adenocarcinoma
 Adenocarcinoma, NOS
 NUT carcinoma
 Undifferentiated carcinoma
 Other rare thymic carcinomas
 Adenosquamous carcinoma
 Hepatoid carcinoma
 Thymic carcinoma, NOS

Thymic neuroendocrine tumors

빈도가 많진 않음

Carcinoid tumors
 Typical carcinoid
 Atypical carcinoid
 Large-cell neuroendocrine carcinoma
 Combined large-cell neuroendocrine carcinoma
 Small-cell carcinoma (SCC)
 Combined SCC

Mediastinal disease

Mediastinal tumor

Thymoma

Epithelial cell과 lymphocyte 어떻게 구성하느냐에 따라 type들이 나뉘지고 있다~정도로만 알면 됨

- M/C anterior mediastinal tumor
- Histologically benign tumor, Malignant potential cancer라고 생각. 일반적인 cancer와 달리 long term F/U (10년 정도)

Thymoma Subtype	Obligatory Criteria	Optional Criteria
Type A	Occurrence of bland, spindle-shaped epithelial cells (at least focally); paucity ^a or absence of immature (TdT+) T cells throughout the tumor	Polygonal epithelial cells; CD20+ epithelial cells
Atypical type A variant	Criteria of type A thymoma; in addition, comedo-type tumor necrosis; increased mitotic count ($>4/2\text{ mm}^2$); nuclear crowding	Polygonal epithelial cells; CD20+ epithelial cells
Type AB	Occurrence of bland, spindle-shaped epithelial cells (at least focally); abundance ^a of immature (TdT+) T cells focally or throughout tumor	Polygonal epithelial cells; CD20+ epithelial cells
Type B1	Thymus-like architecture and cytology: abundance of immature T cells, areas of medullary differentiation (medullary islands); paucity of polygonal or dendritic epithelial cells without clustering (i.e., <3 contiguous epithelial cells)	Hassall's corpuscles; perivascular spaces
Type B2	Increased numbers of single or clustered polygonal or dendritic epithelial cells intermingled with abundant immature T cells	Medullary islands; Hassall's corpuscles; perivascular spaces
Type B3	Sheets of polygonal slightly to moderately atypical epithelial cells; absent or rare intercellular bridges; paucity or absence of intermingled TdT+ T cells	Hassall's corpuscles; perivascular spaces
MNT ^b	Nodules of bland spindle or oval epithelial cells surrounded by an epithelial cell-free lymphoid stroma	Lymphoid follicles; monoclonal B cells and/or plasma cells (rare)
Metaplastic thymoma	Biphasic tumor composed of solid areas of epithelial cells in a background of bland-looking spindle cells; absence of immature T cells	Pleomorphism of epithelial cells; actin, keratin, or EMA-positive spindle cells
Rare others ^b		

Thymoma

- **Clinical presentation**

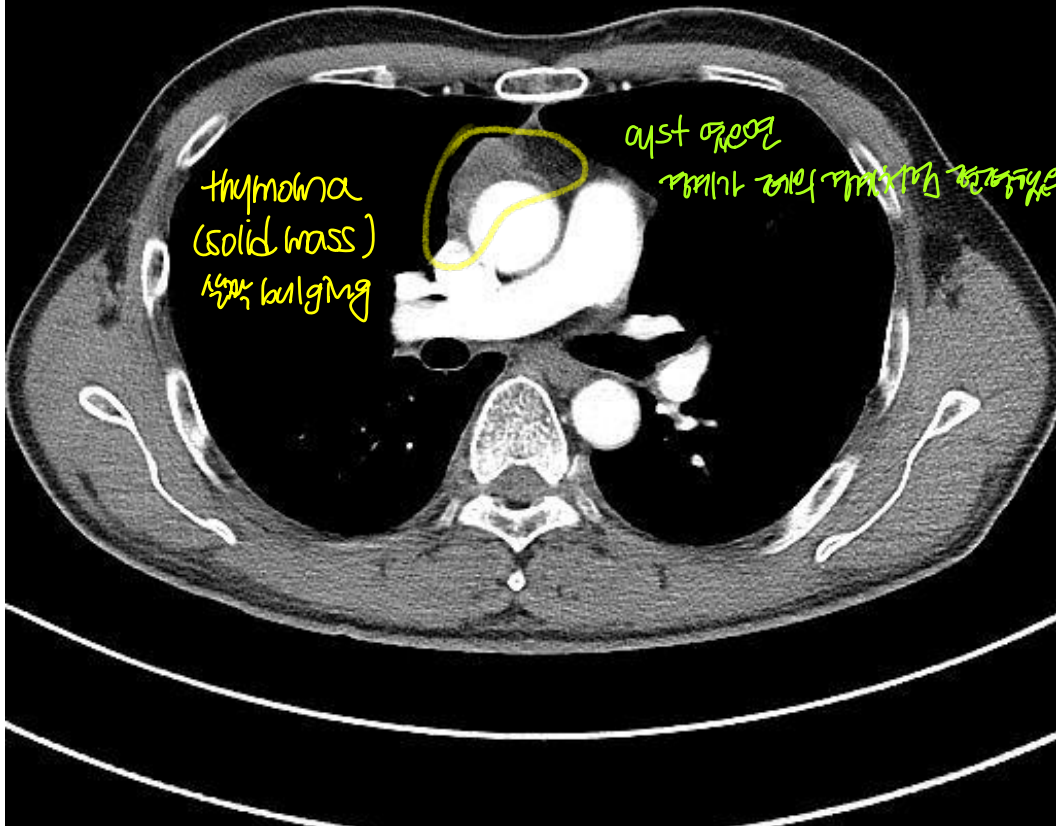
- Older than 40yrs
- Incidental discovered mediastinal mass with asymptomatic patient : 50%
- Local symptom of compression or pain from large mediastinal tumor
 - cyst든 mass든 너무 커지면 mass effect로 압력받아 증상 나타날 수 있음
Vague chest pain, shortness of breath, cough
 - Severe chest pain, SVC obstruction, diaphragm paralysis (phrenic n.), hoarseness (recurrent laryngeal n.)
- Initially symptom related to myasthenia gravis
 - 5-15% patient with MG : found thymoma
 - 30-50% thymoma associated with clinical MG

신경말단에서 연결 통로를 auto-Ab가 막고 있어
전달이 제대로 이루어지지 않아 근육이 빠짐

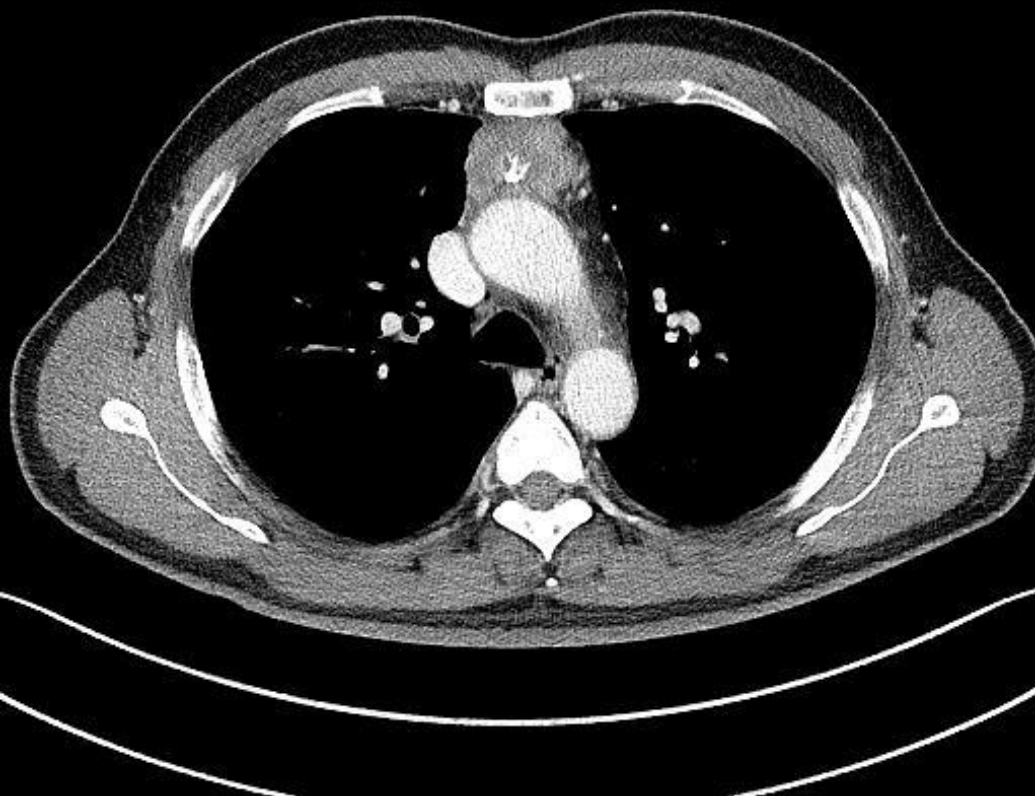
-MG 있으면 5-15% 가 thymoma

-Thymoma 있는 사람 중 30-50%정도가 MG가 있다고 알려져 있음

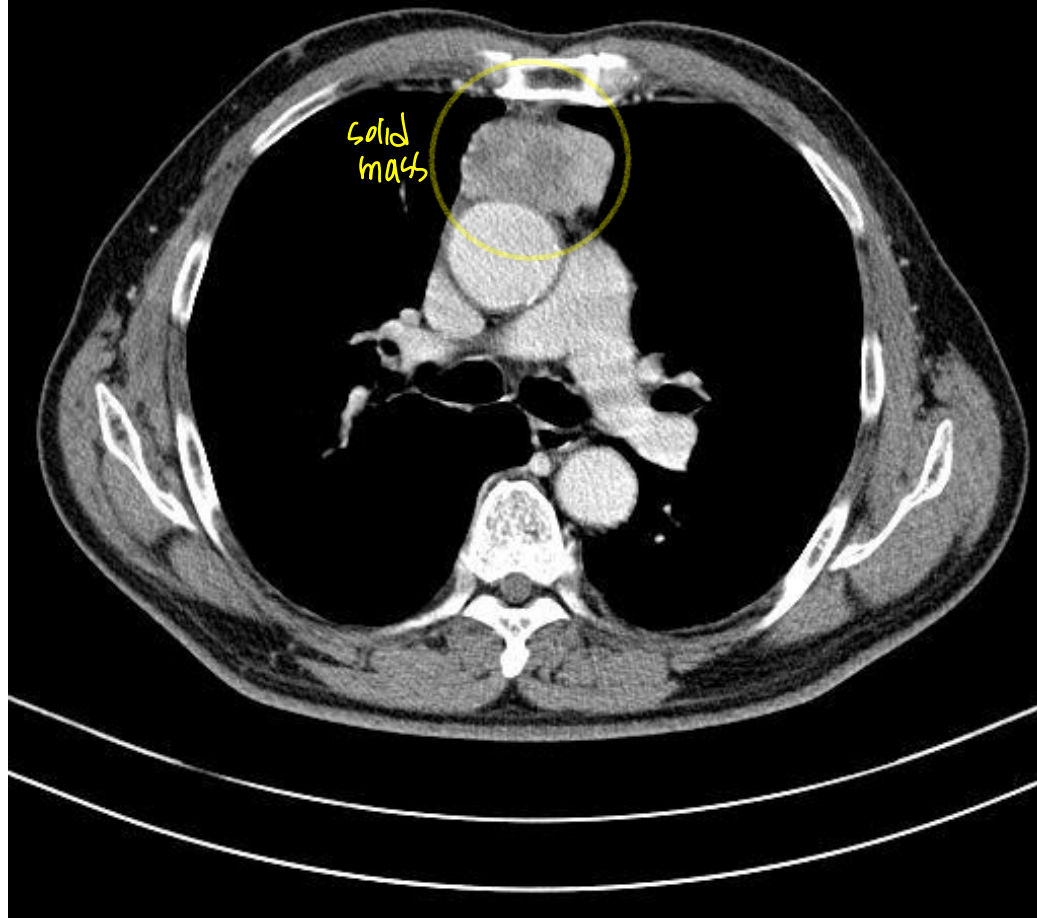
(cyst) 과형성(괴사) → 폐가늠어내는 정도(괴사) 양상



calcification 포함
thyroid carcinoma 의심



solid mass

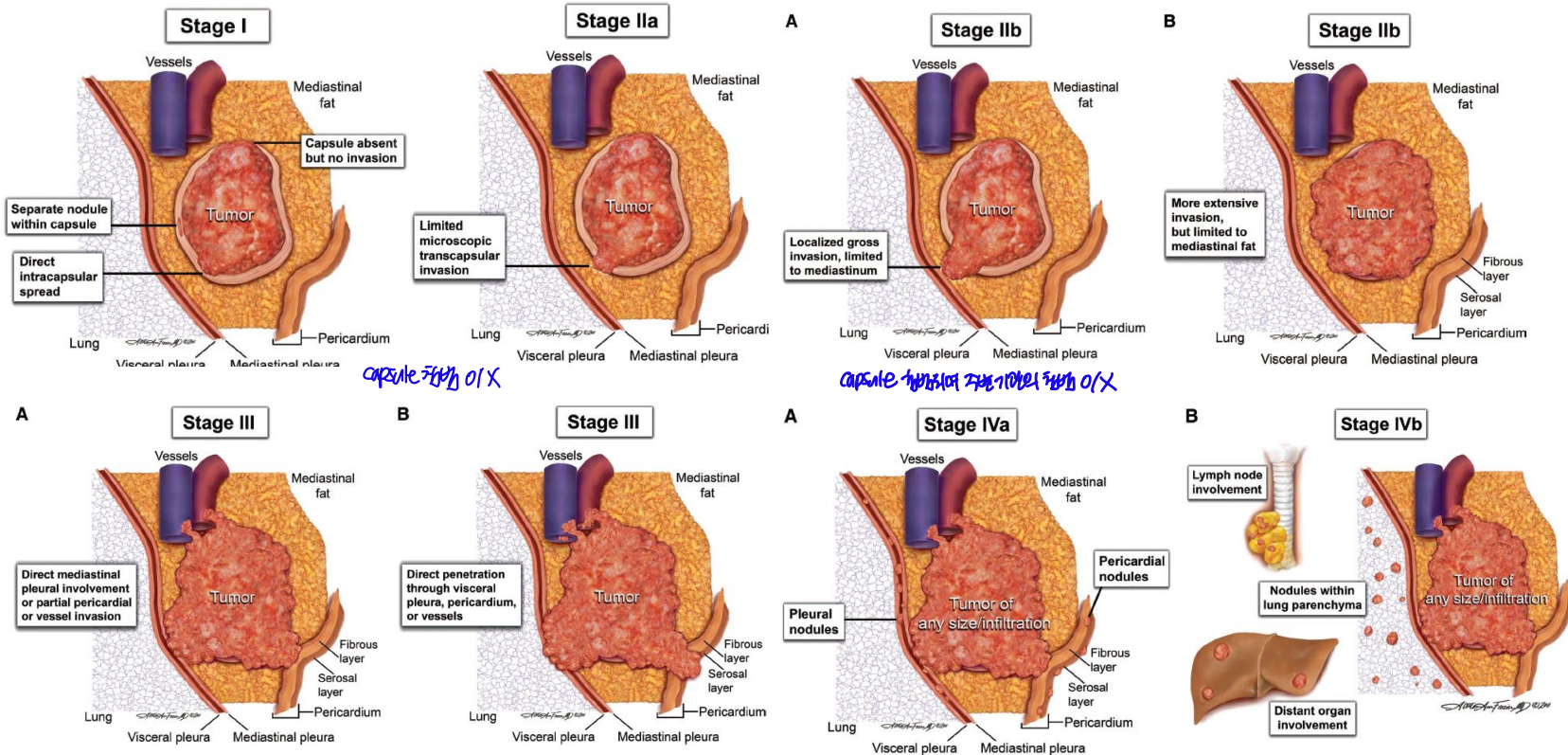


Staging of thymic tumor

- Masaoka-Koga staging system

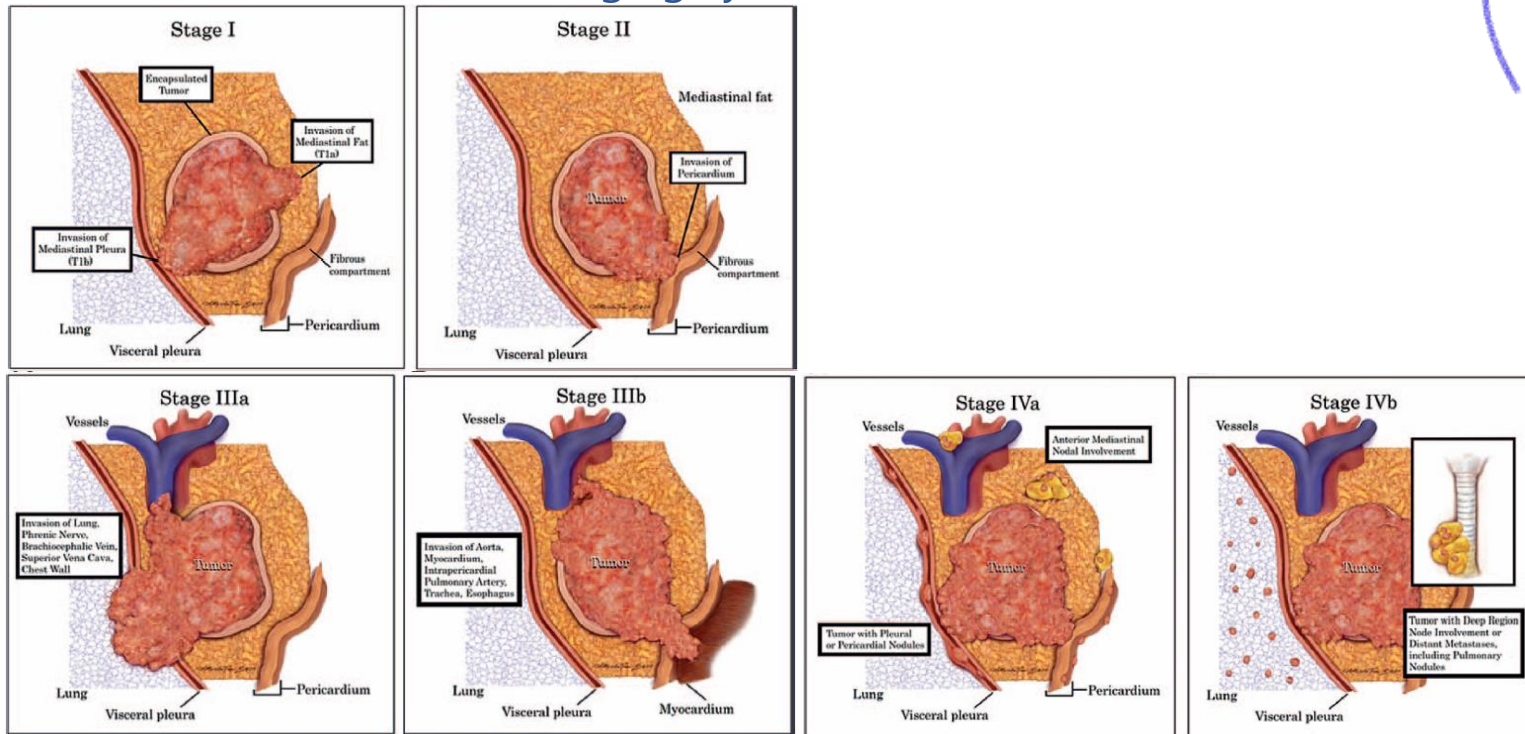
TABLE 1. Masaoka-Koga Staging System

Stage	Definition
I	Grossly and microscopically completely encapsulated tumor
IIa	Microscopic transcapsular invasion
b	Macroscopic invasion into thymic or surrounding fatty tissue, or grossly adherent to but not breaking through mediastinal pleura or pericardium
III	Macroscopic invasion into neighboring organ (i.e., pericardium, great vessel, or lung)
IVa	Pleural or pericardial metastases
b	Lymphogenous or hematogenous metastasis



Staging of thymic tumor 대략적으로 알기

• ITMIG/IASLC TNM staging system



Category	Definition (Involvement of) ^{a,b}
T1	
a	Encapsulated or unencapsulated, with or without extension into mediastinal fat
b	Extension into mediastinal pleura
T2	Pericardium
T3	Lung, brachiocephalic vein, superior vena cava, chest wall, phrenic nerve, hilar (extrapericardial) pulmonary vessels
T4	Aorta, arch vessels, main pulmonary artery, myocardium, trachea, or esophagus

Category	Definition (Involvement of) ^a
N0	No nodal involvement
N1	Anterior (perithymic) nodes
N2	Deep intrathoracic or cervical nodes
M0	No metastatic pleural, pericardial, or distant sites
M1	
a	Separate pleural or pericardial nodule(s)
b	Pulmonary intraparenchymal nodule or distant organ metastasis

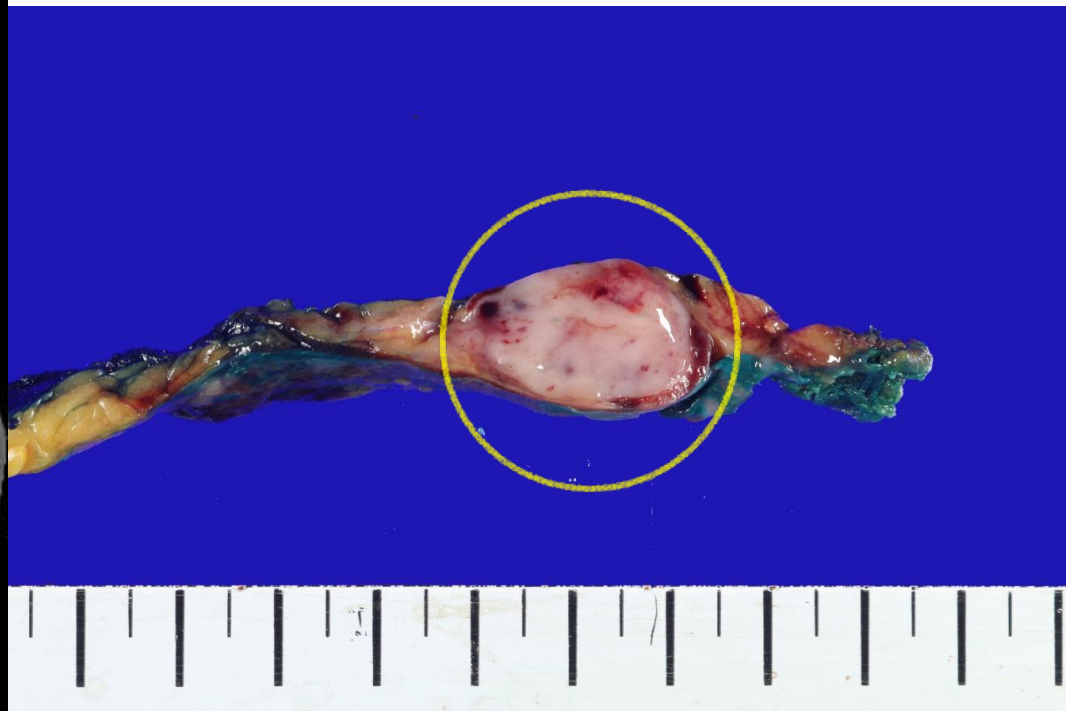
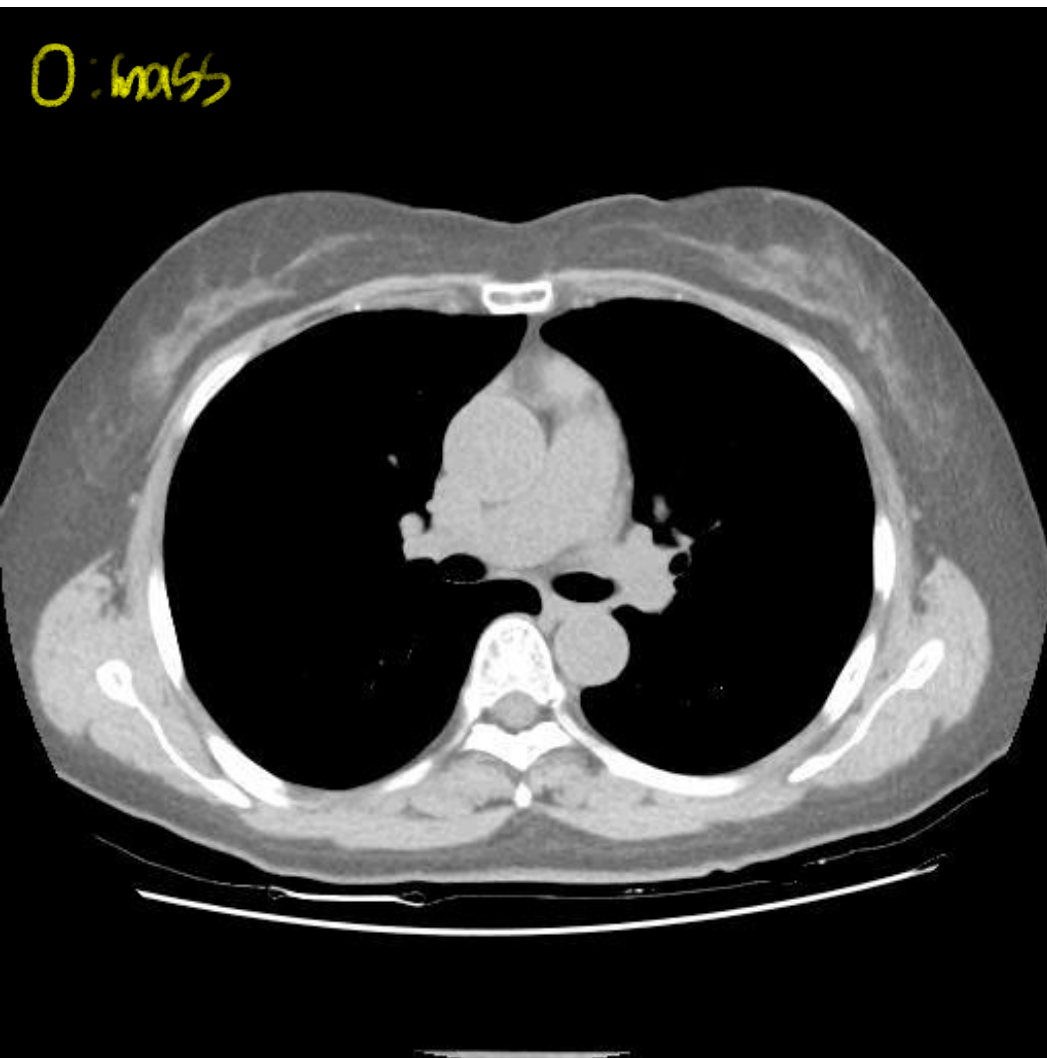
Stage	T	N	M
I	T1	N0	M0
II	T2	N0	M0
IIIa	T3	N0	M0
IIIb	T4	N0	M0
IVa	T any	N1	M0
	T any	N0,1	M1a
IVb	T any	N2	M0,1a
	T any	N any	M1b

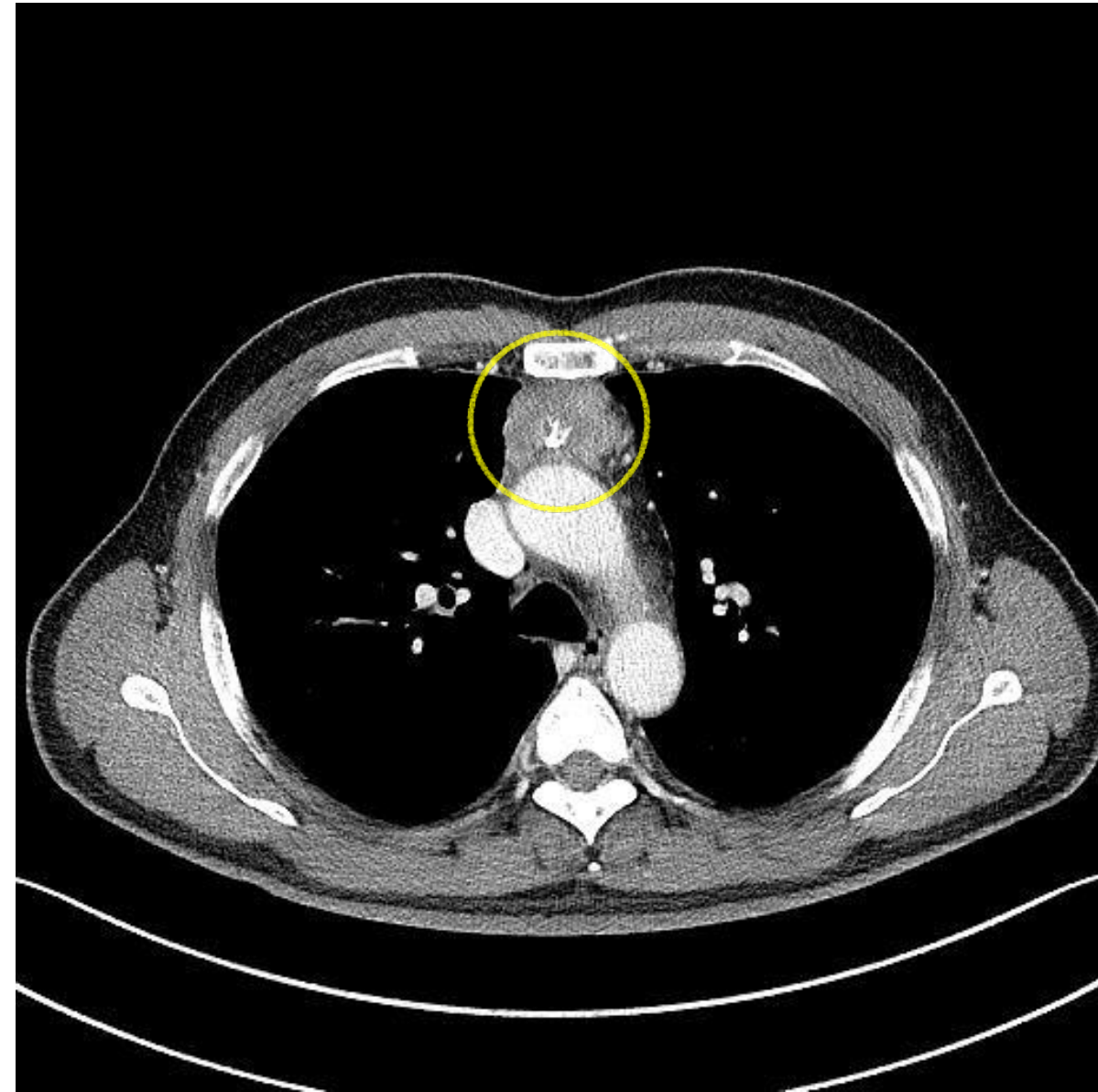
Thymoma

- **Treatment**
- **Early stage tumor : Stage I, II** 다른 질환보다 수술적 처치가 많이 중요
 - Complete **surgical resection** : **R0** resection without violating tumor capsule
 - Total **thymectomy** with en bloc resection tumor with entire thymus gland and perithymic fat
- **Locally advanced tumor**
 - Generally considered surgical disease (except stage IVB)
 - **Neoadjuvant therapy** : improve rate of **R0** resection
 - **Resection margin close or positive**
 - **Postoperative radiation therapy (PORT)**
 - **Thymic carcinoma** benefit from PORT

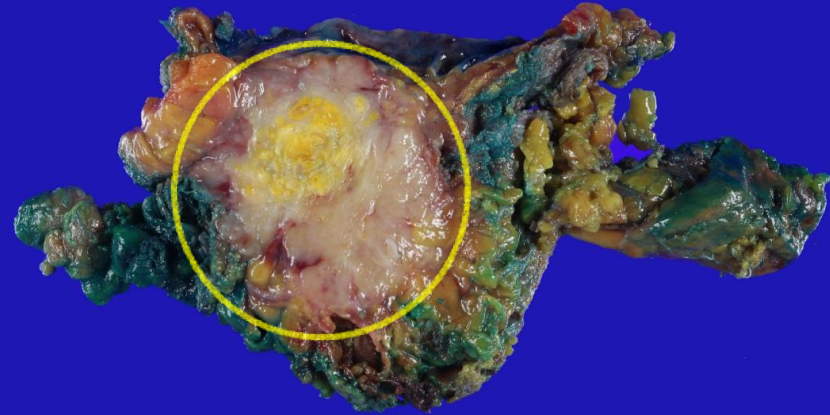
R0: 눈에 안 보이고 microscopic하게 침범X
R1: 눈에는 안 보이는데 microscopic하게 침범
R2: 눈에도 보임

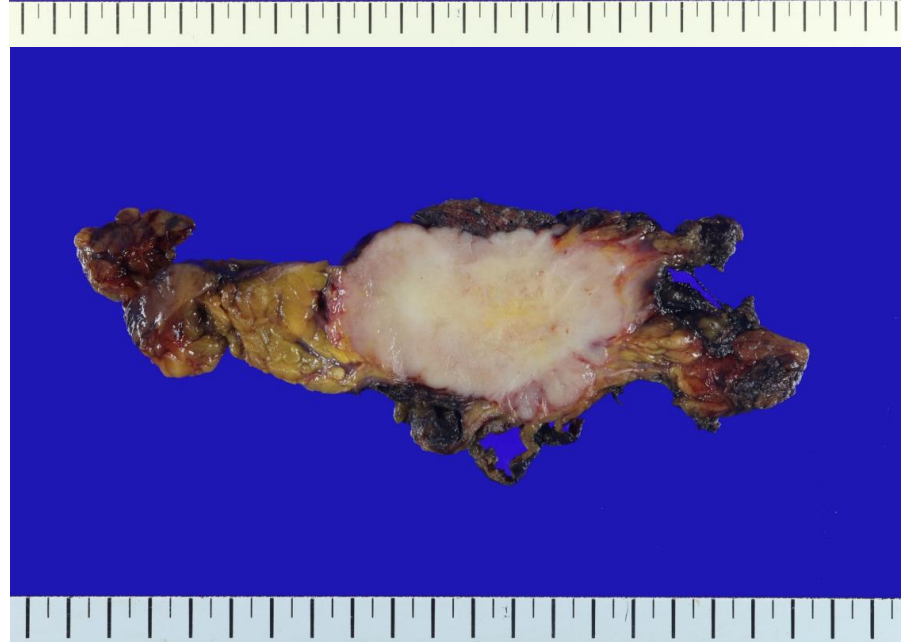
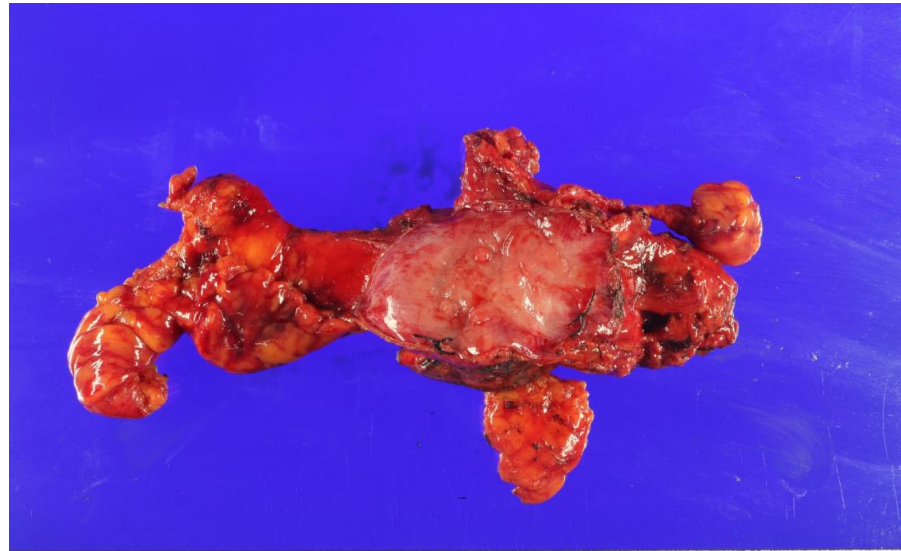
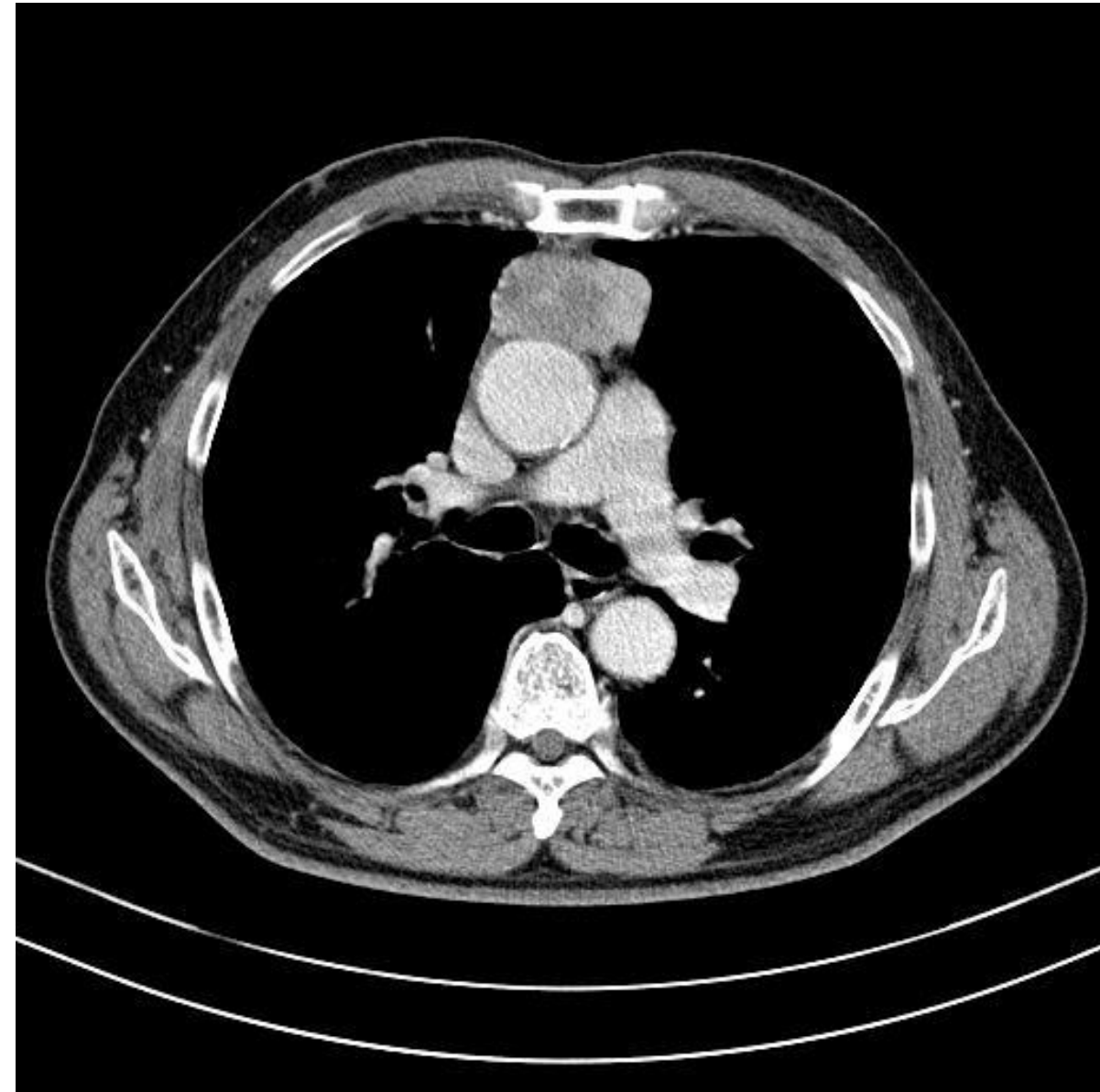
아직까지 chemoRT 하는 경우도 있음





mass+thymus
thymectomy



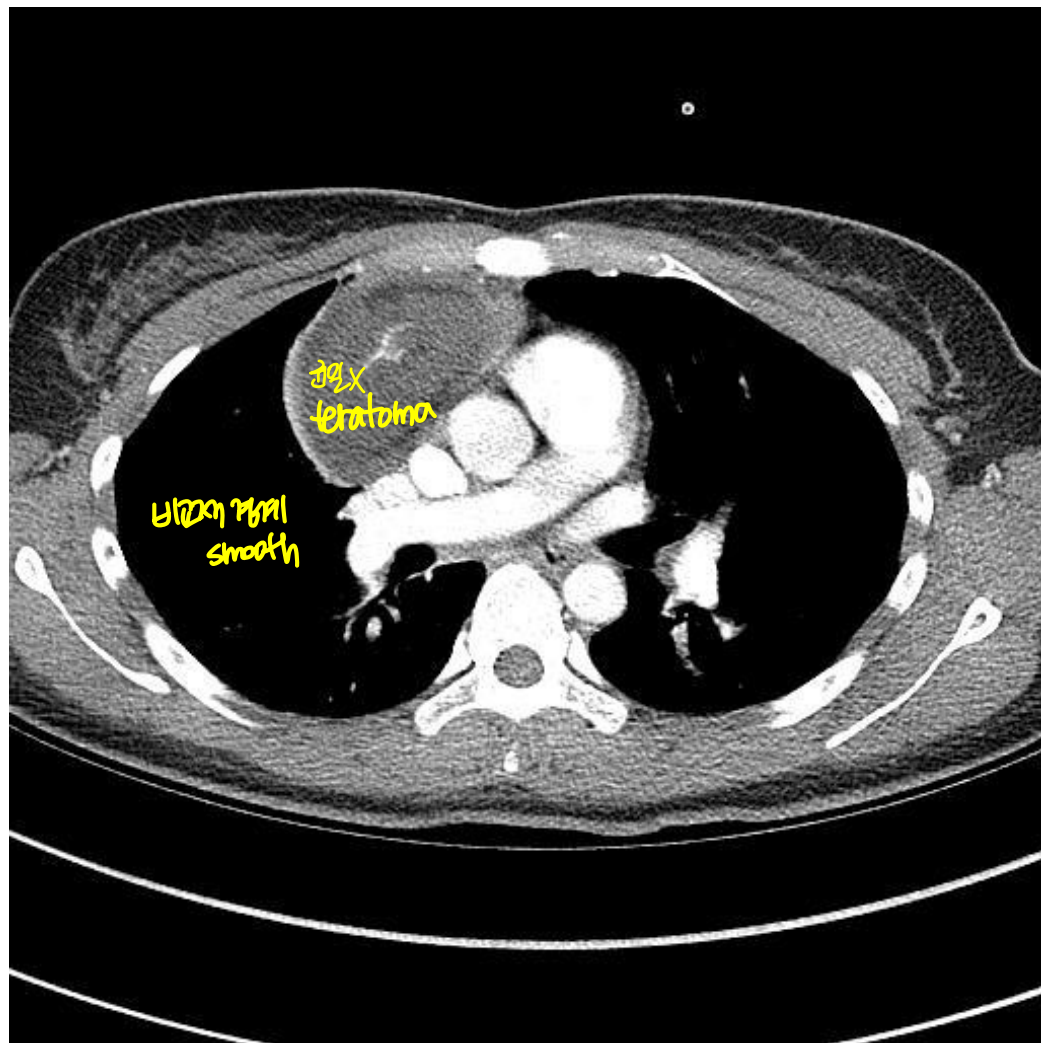


Germ cell tumor

- Group of neoplasm arise in gonadal tissue
- Anterior mediastinum : m/c extragonadal location for GCT
- **Benign Mediastinal Teratoma**
 - Asymptomatic, cough, chest pain, air compression in children 모호한 증상이 많음
 - CT : well circumscribed, variable enhancement (presence of different tissue types including fat, muscle, bone, and cystic components)
 - Treatment : complete resection
- **Malignant Mediastinal Germ Cell Tumor**
 - Seminomas (40%)
 - Nonseminomas (60%) : embryonal cell carcinoma, choriocarcinoma, yolk sac tumors, teratocarcinomas
 - Invade surrounding structure : cause symptom
 - Tumor marker : AFP, β -hCG, LDH
 - pure seminoma : normal AFP, elevated β -hCG
 - pure nonseminoma
 - Elevated AFP : associated with yolk sac tumors
 - Elevated hCG : associated with choriocarcinomas
 - Treatment : chemotherapy, surgical resection

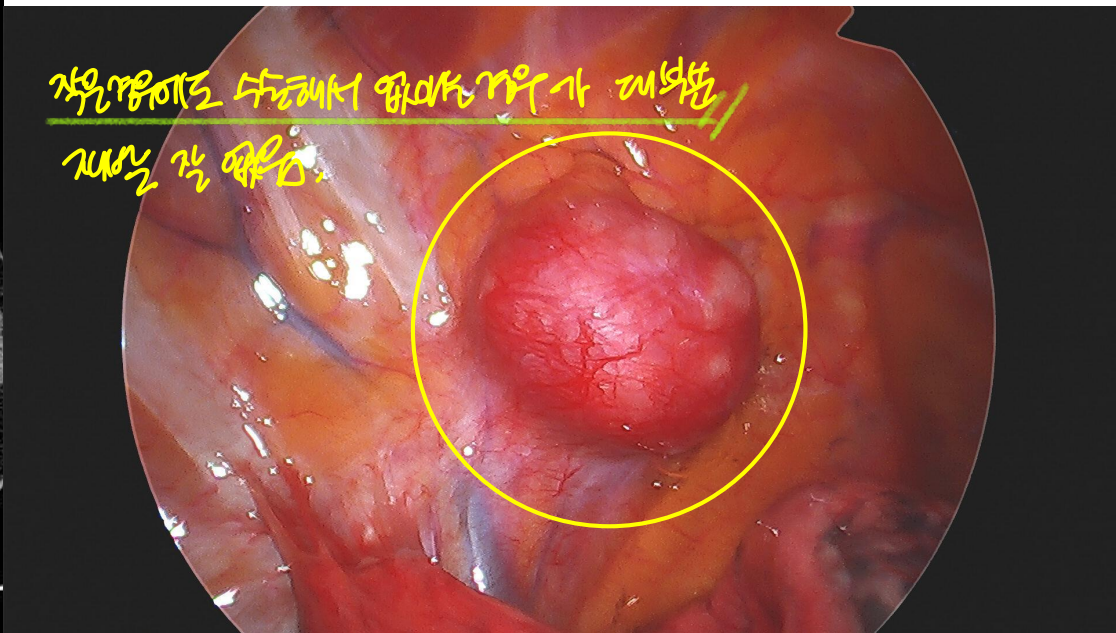
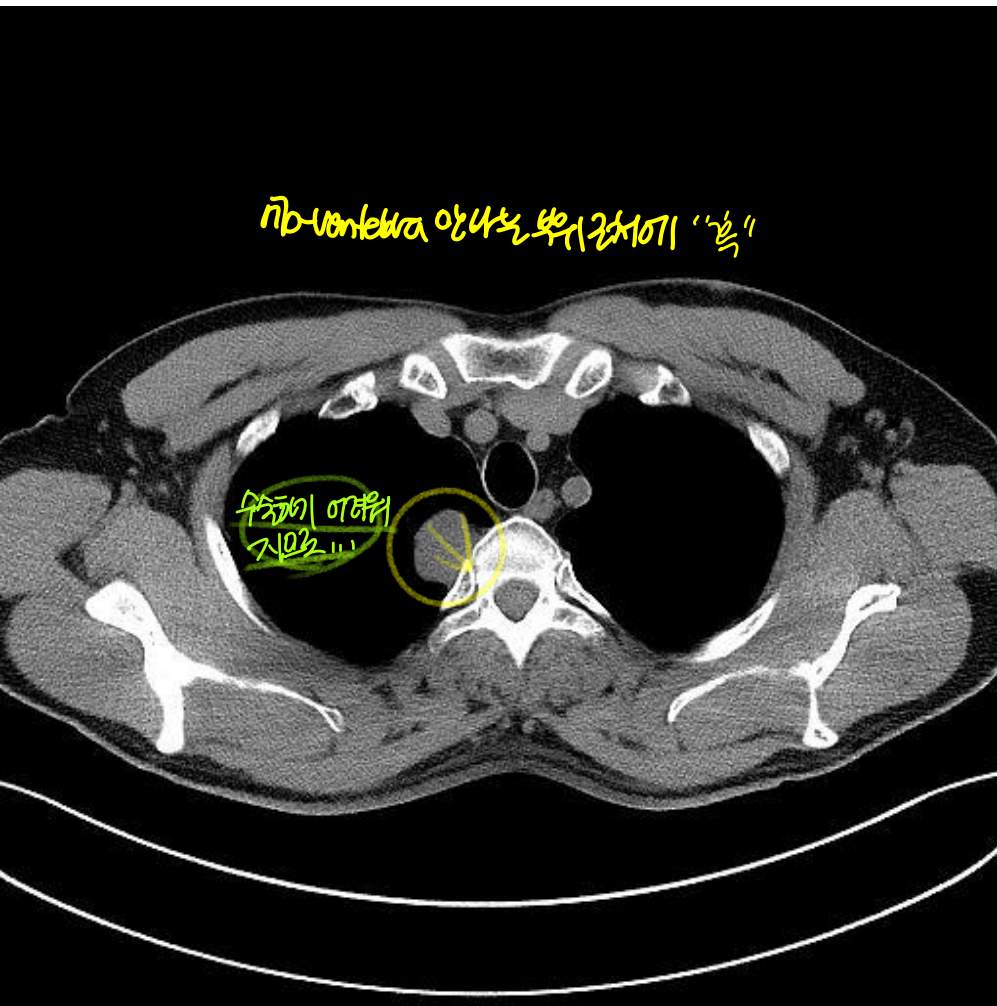
Histologic Diagnosis	Therapeutic Implication
Seminoma	Chemotherapy
Malignant nonseminomatous GCT ^a	Chemotherapy followed by resection of tumor remnants, irrespective of the histology
Mature teratoma	Resection
Immature teratoma	Surgical resection

Tumor	Serum or Urine Marker
Endodermic or yolk sac	α -FP, LDH
Embryogenic	LDH, β -HCG (+/-)
Choriocarcinoma	β -HCG, LDH
Seminoma	β -HCG ^a , LDH, PLAP, NSE



Neurogenic tumor

- Intercostal nerve tumors ^{4/5}
 - Neurofibroma : benign
 - Neurilemoma (schwannoma) : benign, m/c
 - Neurofibrosarcoma
 - Neurosarcoma
- Sympathetic ganglia tumors
 - Ganglioneuroma : benign, m/c in children, paravertebral region
 - Ganglioneuroblastoma : intermediate malignant potential
 - Neuroblastoma : highly malignant, young < 4yrs, highly invasive, metastasis
- Paraganglia cell tumors
 - Paraganglioma (pheochromocytoma)





Bronchogenic cyst 뒤쪽에 많이 생김

- Mediastinal cysts : 20% of all mediastinal mass
- Bronchogenic cysts : 60% of all mediastinal cyst
- Part of the spectrum of bronchopulmonary foregut abnormality
 - Abnormal budding of the tracheolaryngeal groove
- Ciliated, columnar respiratory epithelium or cuboid epithelium containing mucous glands
- Communication with tracheobronchial tree : rare, recurrent infection
- Symptom : compression of adjacent structure, recurrent infection 없는 경우가 대부분이긴 함
- Surgical resection
 - Question of malignant transformation, positive results of cytology, evidence of enlargement or recurrence, presence of symptoms, communication with bronchial tree

